these are all the files, take your time, explain everything the best you could, both the theory and practical

Perfect — I've read through all 7 files thoroughly. Here is your **complete study guide** for the midterm, based exactly on your professor's slides and exercise sheets. The exam covers **Chapters 1–4** only (no compression yet).

Chapter 1 — Introduction to Multimedia

What is Multimedia?

Multimedia is the combination of different media types (text, audio, image, video, graphics) into a digital system. The word appeared in the early 1990s, after the success of digital audio on CDs, followed by the first multimedia CD-ROMs.[1]

Types of Multimedia Data

Your professor lists these as the **mainstream** types:[1]

- **Text** — formatted or hypertext (non-linear, e.g. web links)
- **Audio** — characterized by dimensionality, frequency range (voice vs. music)
- **Image** — 2D array of pixels, has bit-depth
- **Video** — aspect ratio, width/height, 1D scanning
- **Graphics** — 2D/3D vector coordinates, color, thickness
- Not-yet-mainstream: **Holograph**, **Haptic**

Key Qualities of Multimedia Data

- **Digital** — always bits and bytes
- **Voluminous** — storage/transmission challenges → need for compression
- **Interactive & Jumping** — e.g. skipping chapters in a PDF
- **Real-time & synchronization** — e.g. watching a live football match [1]

Classification of Multimedia Systems

Dimension	Category A	Category B
Temporality	Static (.ppt)	Dynamic (video)
Timing	Real-time (live broadcast)	Orchestrated (DVD)
Structure	Linear (eBook)	Non-linear (hypertext)
Interaction	Person-to-machine (game)	Person-to-person (videoconference)
Distribution	Single user	Peer-to-peer / Broadcast

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Standardization Bodies to Know

- **ISO** — International Organization for Standardization
- **ITU** — International Telecommunication Union
- **JPEG** — Joint Pictures Expert Group
- **MPEG** — Motion Pictures Expert Group
- **OSI** — Open Systems for Interconnection [1]

⚡ Chapter 2 — Digital Data Acquisition

This is the most math-heavy chapter and has the most practical problems.

Analog vs. Digital

An **analog signal** is a continuous function (infinite possible values). A **digital signal** is discrete — it has been sampled and quantized. The advantages of going digital: no degradation over time, easy to compress, easy to store and transmit.[2]

ADC: Sampling + Quantization

Converting analog to digital has two steps:

1. Sampling — measuring the signal at regular time intervals. The result is a discrete time series.

2. Quantization — rounding each measurement to the nearest allowed level. The key formula is:

$$\Delta = \frac{R}{N} = \frac{R}{2^b}$$

Where Δ is the **quantization step**, R is the **voltage range** (max − min), b is the number of **bits**, and $N = 2^b$ is the number of levels.[3]

The **quantization error** can never exceed half the step size:

$$e_{max} = \frac{\Delta}{2} = \frac{R}{2N} = \frac{R}{2^{b+1}}$$

The Nyquist Theorem

Established by **Henry Nyquist (1920)** and **Claude Shannon (1950)**:[2]

> A signal must be sampled at a frequency **greater than twice** the maximum frequency in the signal.

$$f_{sample} > 2 \times f_{max}$$

The value $2 \times f_{max}$ is called the **Nyquist frequency**. For example: human hearing goes up to ~20 kHz → CDs use 44.1 kHz (just above 40 kHz Nyquist rate).[2]

Aliasing

Aliasing = loss of information when the sampling rate is **lower** than the Nyquist frequency. The signal cannot be correctly reconstructed. A classic visual example: car wheels appearing to spin backward on film.[2]

Over-sampling (above Nyquist) is fine — it just wastes storage and bandwidth.

Bit Rate

$$\text{Bit Rate} = f_{sample} \times b$$

Where f_{sample} is sampling frequency (Hz) and b is bits per sample.[3]

Fourier Transform

Breaks any signal into its component frequencies (sine waves). Used in filtering and understanding what frequencies a signal contains. LTI (Linear Time-Invariant) systems respond to each frequency independently — this makes Fourier analysis powerful.[2]

⊠ Chapter 3 — Media Representations & Formats

Digital Images

An image is a 2D array of pixels. Its raw (uncompressed) size is calculated as:[4]

$$\text{Size (bits)} = \text{width} \times \text{height} \times \text{bit depth}$$

Image type	Bit depth
Grayscale	8 bits/pixel (1 channel)
Color (RGB)	24 bits/pixel (3 channels × 8 bits)
With alpha channel	32 bits/pixel
Video game typical	16 bits/pixel (5R + 5G + 5B + 1 alpha)

Example from slides: A 640×480 color image → 640 × 480 × 3 × 8 = 7.37 Mbits = **921.6 KB**.[4]

Aspect Ratios

Ratio	Use case
3:2	Standard printed photographs
4:3	Traditional TV
16:9	HD / modern TV
47:20	Cinema (anamorphic)

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Image Formats

- .BMP — raw uncompressed bitmap, best for editing
- .GIF / .PNG — lossless, best for poster-like images with solid colors
- .JPG — lossy, best for photographs
- **Vector formats (SVG, CGM)** — stored as shapes/coordinates, not pixels; infinitely scalable [4]
- **Metafiles** — combine vector + bitmap (e.g. CGM, SVG, WebCGM) [4]

Digital Video

Video = a sequence of **frames** shown in rapid succession. Key properties:[4]

- **Frame rate**: Film = 24fps, TV (NTSC) = 30fps, TV (PAL) = 25fps
- **Flicker** occurs when frame rate is too low [4]
- **Interlaced scanning**: Each frame is split into two fields (odd + even lines), transmitted alternately. **Solves flicker** at lower bandwidth [4]
- **Progressive scanning**: All lines sent in sequence — cleaner for fast motion

YUV Color Space & Subsampling

The **YUV** (or YCbCr) format separates **Y** (luminance = brightness) from **U and V** (chrominance = color). Human eyes are far more sensitive to brightness than color, so you can throw away color detail without noticeable quality loss.[4]

The notation **4:X:Y** describes how often color channels are sampled relative to luma per 2×2 block:

Scheme	Bits/pixel	Description
4:4:4	24	Full color, no subsampling
4:2:2	16	U and V sampled every 2 pixels horizontally
4:2:0	12	U and V sampled every 2 pixels both horizontally and vertically
4:1:1	12	U and V sampled every 4 pixels horizontally

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How to prove 4:2:0 = 12 bits/pixel: In a 2×2 block of 4 pixels, you have 4 Y samples (4×8=32 bits), 1 U sample (8 bits), 1 V sample (8 bits) → total 48 bits ÷ 4 pixels = **12 bits/pixel.**[5]

Digital Audio

Sound is pressure waves; key properties to know:[4]

- **Amplitude** → loudness (measured in **decibels**)
- **Frequency** → pitch (measured in **Hz**, cycles per second)
- **Period** → time between two crests
- **Bandwidth** = highest frequency − lowest frequency
- **Timbre** = the combination of fundamental + harmonics (overtones); this is why a violin and flute sound different on the same note
- Human hearing range: **20 Hz – 20,000 Hz**, most sensitive at **2–4 kHz**[4]

Audio File Size Formula

$$\text{File size (bytes)} = f_{sample} \times t \times \frac{b}{8} \times \text{channels}$$

Where t is duration in seconds, b is bits per sample, channels = 1 (mono) or 2 (stereo).[4]

Example from slides: CD quality stereo → 44,100 × 1s × (16/8) × 2 = **176,400 bytes = 172 KB per second.**

⊠ Chapter 4 — Color Theory

Human Vision

The retina has two types of photoreceptors:[6]

- **Rods** (~120 million) — detect light/dark, uniformly distributed, very sensitive
- **Cones** (~7 million) — detect color (R, G, B types), concentrated near the **fovea**, less sensitive than rods

The **Young-Helmholtz theory** states that color perception requires three types of receptors (R, G, B) — this is called **trichromacy**. Most mammals only have two (dichromacy). Marine mammals have one (monochromacy).[6]

Cone Response

Each cone type has a different **spectral absorption curve**. The energy response is the convolution of the cone's response function $s(\lambda)$ with the spectral distribution of the light $f(\lambda)$. The result is the **tristimulus vector** — three numbers (one per cone type) that represent any perceived color.[6]

Color Spaces

Space	Description	Use
RGB	Additive: Red + Green + Blue	Screens/displays
CMY/CMYK	Subtractive; CMY = 1 − RGB; K = black ink	Printing
YUV/YCbCr	Luminance Y + two chrominance channels	Video/broadcasting
HSV	Hue, Saturation, Value	Intuitive color editing
XYZ	CIE standard reference space	Colorimetry

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CMY ↔ RGB conversion (important!):

$$C = 1 - R, \quad M = 1 - G, \quad Y = 1 - B$$

CCD Cameras

A CCD (Charge-Coupled Device) is a rectangular grid on silicon. Photons hitting the silicon release charge proportional to light energy. The sensing element itself is **monochromatic** — color is captured using color filters over the sensor.[6]

⊠ Practical Problems — Fully Worked

Type 1: ADC / Quantization

Problem 1 (from your Zadaci.docx): 8-bit ADC, range –2V to +4V. Detected hex value = A2.[3]

Step 1: Convert A2 hex → decimal: A=10, so A2 = 10×16 + 2 = **162**

Step 2: Voltage range R = 4 – (–2) = **6V**

Step 3: Quantization step: $\Delta = 6/2^8 = 6/256 = 0.02344$ V

a) Input voltage = 162 × 0.02344 = 3.797V (this is offset from 0), then subtract min: 3.797 – 2 = **1.797 V** ✓

b) Max quantization error = $\Delta/2 = 0.02344/2 = \mathbf{0.01172\ V}$

c) With 12 bits: $\Delta = 6/4096 = 0.001465$ V, error = **0.000732 V** ✓ [3]

Type 2: Nyquist & Bit Rate

Problem 2: Speech: 50Hz–10kHz, 12 bits/sample. Music: 15Hz–20kHz, 16 bits/sample.[3]

a) Speech Nyquist rate = 2 × 10,000 = **20 kHz**. Bit rate = 20,000 × 12 = **240 kbps**

b) Music Nyquist rate = 2 × 20,000 = **40 kHz**. Bit rate = 40,000 × 16 = **640 kbps**

c) 10 min stereo music: 640,000 × 600s × 2 channels = 768,000,000 bits = **93.75 MB** ✓ [3]

Type 3: Image Size

Problem 3: Photo 5×7 inches, 120 dpi, 24-bit color.[3]

Pixels = (5×120) × (7×120) = 600 × 840 = **504,000 pixels**

a) 24-bit color = 3 bytes/pixel → 504,000 × 3 = **1,512,000 bytes ≈ 1.44 MB**

b) 256 colors = 8-bit = 1 byte/pixel → 504,000 × 1 = **504,000 bytes = 0.48 MB**[3]

Type 4: Aliasing (The Car Wheels Problem)

Problem 5: Car at 36 km/h, wheel diameter = 0.4244m, filmed at 24fps.[3]

- Speed = 36 km/h = **10 m/s**
- Wheel circumference = π × d = π × 0.4244 = **1.333 m**
- Wheel rotation frequency = 10 / 1.333 = **7.5 Hz**
- **a)** Nyquist rate = 2 × 7.5 = **15 Hz**
- **b)** 24fps > 15Hz → wheels appear to spin at **real speed** ✓
- **c)** 12fps < 15Hz → aliasing! Apparent frequency = 7.5 − 12 = **−4.5 Hz** (backward!); at 7.5fps: 7.5 − 7.5 = **0 Hz** (frozen!); at 6fps: 7.5 − 6 = **1.5 Hz** (slowly forward)
- **d)** Max speed before aliasing at 24fps: Nyquist = 12Hz. Ratio: 7.5/12 = 36/x → x = **57.6 km/h**[3]

Type 5: YUV Subsampling Bit Rate

Problem 2 (Subsampling): Camera: 450 lines, 520 pixels/line, 25fps, 4:2:0, 8 bits/channel.[5]

In **4:2:0**: per 2×2 block → 4 Y samples + 1 Cb + 1 Cr. Effective bits/pixel = 12.

- Total pixels/frame = 450 × 520 = 234,000
- Bits/frame = 234,000 × 12 = 2,808,000 bits
- **Bit rate** = 2,808,000 × 25 = **70.2 Mbps**

For 10 minutes storage with Cb/Cr requantized to 6 bits (so now 4+1+1 channels at 8,8,6,6 effectively → 8×4 + 6 + 6 = 44 bits per 4 pixels = 11 bits/pixel):

- Bits = 11 × 234,000 × 25 × 600 = **38,610,000,000 bits ≈ 4.5 GB**[5]

Type 6: Compression Rate for Transmission

Problem 4 (Subsampling): 620×480 video, interlaced, 59.94 Hz field rate, 4:2:2, send over 1.5 Mbps. [5]

- Frame rate = 59.94/2 = 29.97 fps (interlaced = 2 fields/frame)
- Pixels/frame = 620 × 480 = 297,600
- 4:2:2 = 16 bits/pixel
- Raw bit rate = 297,600 × 16 × 29.97 = **~142.7 Mbps**
- Compression ratio needed = 142.7 / 1.5 = **~95:1**

For 4:1:1 (also 12 bits/pixel):

- Raw bit rate = 297,600 × 12 × 29.97 = **~107 Mbps** → ratio = **~71:1** [5]

⊠ Exam Quick-Reference

Based on the course info, your midterm covers **Chapters 1–4**. The most likely question types: [7]

Question type	Formula to remember
Quantization step	$\Delta = R/2^b$
Quantization error	$e = \Delta/2$
Nyquist rate	$f_s = 2 \times f_{max}$
Audio bit rate	$f_s \times b_{persample}$
Audio file size	$f_s \times t \times (b/8) \times \text{channels}$
Image size (bits)	$W \times H \times \text{bit depth}$
4:2:0 bits/pixel	12 (prove via 2×2 block: 4Y+1U+1V = 48 bits / 4 px)
4:2:2 bits/pixel	16
Aliasing apparent freq	$f_{apparent} = f_{signal} - f_{sample}$

Let me know if you want me to go deeper on any specific topic, create extra practice problems, or explain any formula step by step!

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1. 1_Intro.pptx
2. 2_Digital_data.pptx
3. Zadaci.docx
4. 3_Media_representations.pptx
5. Subsampling_Zadaci.docx
6. 4_Color_Theory-2.pptx

7.0_Info.pptx